

2. Random Variables (Part II)

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Deep Generative Models: A Statistical Perspective



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Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Outline

1. Random Variables
2. Discrete and Continuous Probability Distributions
3. Summary Statistics and Independence
4. Multivariate Gaussian Distributions
5. Conditional Distributions and Bayesian Networks

Conditional Distribution

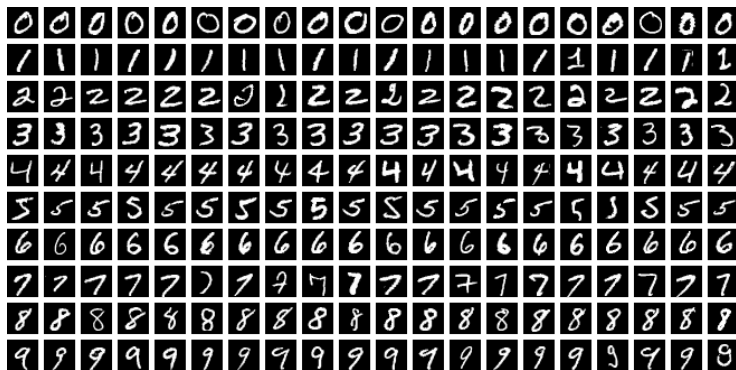
- ▶ In probabilistic modeling, we often seek to understand how the distribution of one random variable changes conditionally on another.
- ▶ For two continuous random variables X and Y , the conditional probability density function of X given $Y = y$ is defined as:

$$f(x | y) = \frac{f(x, y)}{f(y)}$$

assuming $f(y) > 0$.

- ▶ X and Y are independent if and only if $f(x | y) = f(x)$ for all x and y .

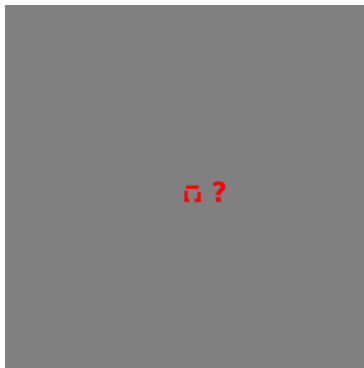
Conditional Distribution



- ▶ Consider images from the Modified National Institute of Standards and Technology (MNIST) database (LeCun, 1998).
- ▶ Each image consists of $28 \times 28 = 784$ binary pixels, with values in $\{0, 1\}$.

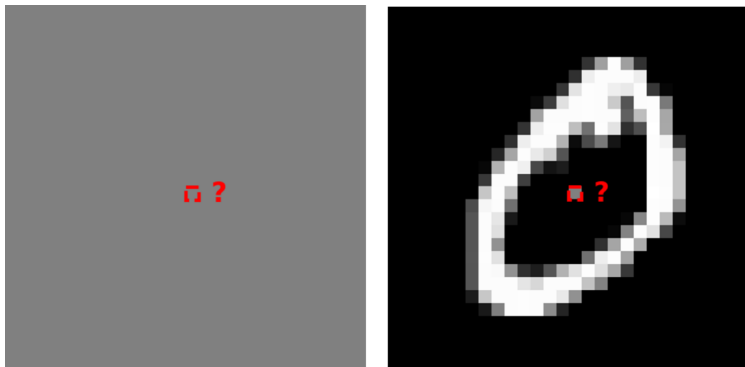
Image source: https://en.wikipedia.org/wiki/MNIST_database.

Conditional Distribution



Outcome (x)	0 (black)	1 (white)
$P(X_{14 \times 14} = x)$	48.08%	51.92%

Conditional Distribution



Outcome (x)	0 (black)	1 (white)
$P(X_{14 \times 14} = x \mid \mathbf{X}_{-14 \times 14})$	48.08% $\rightarrow$?	51.92% $\rightarrow$?

Bayesian Network

- ▶ The *chain rule* can factor any joint PDF $f_{\mathbf{X}}(\mathbf{x})$ of a random vector $\mathbf{X} = (X_1, \dots, X_d)^T$ into a product of conditional densities.
- ▶ For example, when $d = 2$,

$$f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1)f_{X_2|X_1}(x_2 | x_1) = f_{X_2}(x_2)f_{X_1|X_2}(x_1 | x_2). \quad (1)$$

- ▶ As another example, when $d = 3$,

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}) &= f_{X_1}(x_1)f_{X_2|X_1}(x_2 | x_1)f_{X_3|X_1, X_2}(x_3 | x_1, x_2) \\ &= \dots \\ &= f_{X_3}(x_3)f_{X_2|X_3}(x_2 | x_3)f_{X_1|X_2, X_3}(x_1 | x_2, x_3). \end{aligned} \quad (2)$$

- ▶ By repeatedly applying the definition of conditional probability:

$$f(x_1, \dots, x_d) = f(x_1) \prod_{i=2}^d f(x_i | x_1, \dots, x_{i-1}).$$

Q: How do we model the dependence within $\mathbf{X}$ or the joint PDF $f(x_1, \dots, x_d)$?

Bayesian Network

1. **Full Model:** Without any domain knowledge, we can model all the conditional distributions,

$$f(x_i \mid x_1, \dots, x_{i-1}). \quad (3)$$

- ▶ It can express any joint distribution, but the model size becomes large.
2. **Degenerate Model:** When $(X_2, \dots, X_d)$ are known deterministic functions of X_1 , modeling $f(X_1)$ alone is sufficient.
- ▶ The model size is small and even invariant to the data dimension. However, the model capacity is significantly reduced.

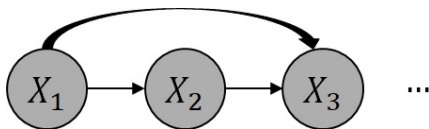
Building appropriate models that reflect domain knowledge is crucial.

Bayesian Network

- ▶ *Bayesian networks* are probabilistic graphical models that represent conditional dependencies using directed acyclic graph structures.
- ▶ Again, we can express the joint PDF $f(x_1, \dots, x_d)$ as

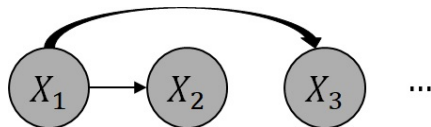
$$f(x_1)f(x_2 \mid x_1) \cdots f(x_d \mid x_1, x_2, \dots, x_{d-1}). \quad (4)$$

Many dependency structures can be represented by a directed graph with d nodes $X_1, \dots, X_d$ and up to $d(d-1)/2$ directed edges.

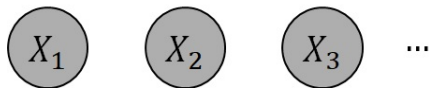


- ▶ For example, **1. Full Model** corresponds to a graph utilizing all nodes and all possible directed edges.

Bayesian Network

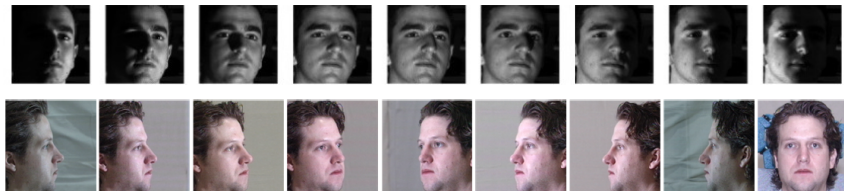


2. **Degenerate Model:** This model is a special case of the above graph where $f(x_1, \dots, x_d) = f(x_1)f(x_2 | x_1) \cdots f(x_d | x_1)$.



3. **Fully Independent Model:** This model assumes mutual independence among all variables, such that $f(x_1, \dots, x_d) = f(x_1) \cdots f(x_d)$.

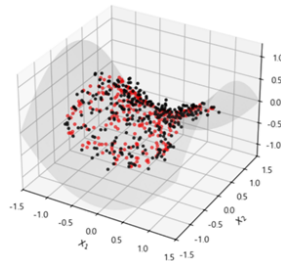
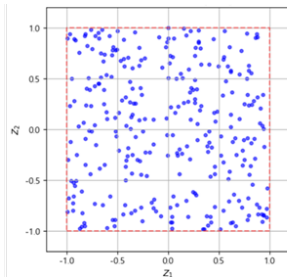
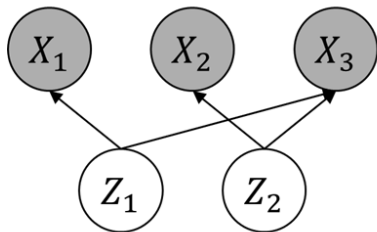
Bayesian Network: Latent Variable Model



4. **Latent Variable Model:** There are latent factors $\mathbf{Z} := (Z_1, \dots, Z_r)^T$, typically consisting of independent components, that are mixed to generate data, e.g., $\mathbf{X} = \mathbf{W}\mathbf{Z}$ in Independent Component Analysis (Hyvärinen and Oja, 2000).
- ▶ The age, light source location, and camera angle are examples of latent factors.

The top and bottom images are from the Extended Yale-B (Georghiades et al., 2001) and Multi-pie (Gross et al., 2010) datasets, respectively.

Bayesian Network: Latent Variable Model



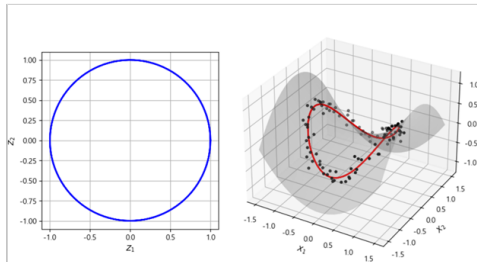
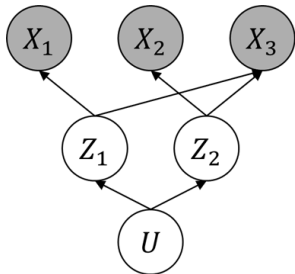
- For example, suppose $\mathbf{Z} \sim \mathcal{U}[-1, 1] \times \mathcal{U}[-1, 1]$ is drawn from a square and $\mathbf{X}$ is:

$$X_1 = Z_1 + \epsilon_1, \quad X_2 = Z_2 + \epsilon_2, \quad X_3 = (Z_1^2 - Z_2^2)/2 + \epsilon_3 \quad (5)$$

where ϵ represents white noise.

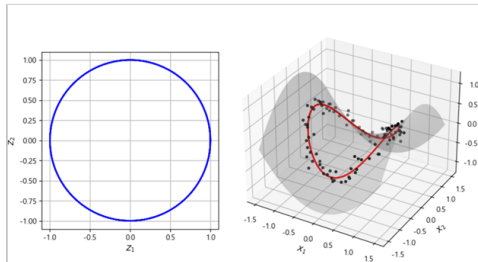
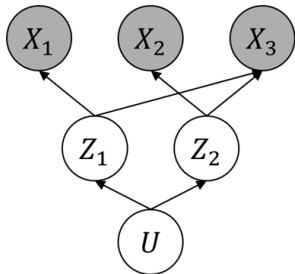
- Here, Z_1 and Z_2 are *parent nodes* with no incoming edges, while X_1 , X_2 , and X_3 are *child nodes* with ones.

Bayesian Network: Latent Variable Model



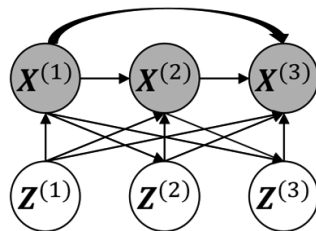
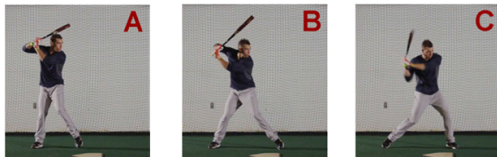
► Now, suppose $\mathbf{Z} = (\cos 2\pi U, \sin 2\pi U)^T$ where $U \sim \mathcal{U}[0, 1]$.

Bayesian Network: Latent Variable Model



- The joint PDF can be expressed as:

Bayesian Network: Latent Variable Model



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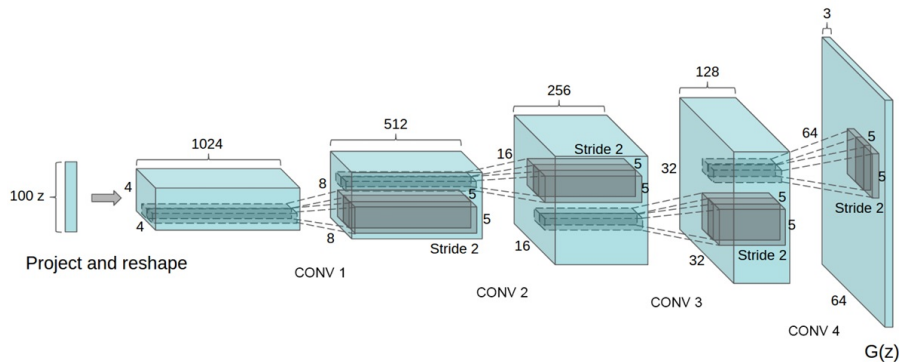
Left images are edited from Szeto et al. (2019).

Bayesian Network: Latent Variable Model

- ▶ Joint PDFs from Bayesian networks can be expressed as the product of conditional densities of each node given its parents, denoted as $\text{Pa}(X_i)$:

$$f(x_1, \dots, x_d) = \prod_{i=1}^d f(x_i \mid \text{Pa}(x_i))$$

Bayesian Network: Latent Variable Model



- ▶ Deep generative models are predominantly based on latent variable models.
- ▶ The conditional distributions $f(\mathbf{X}|\mathbf{Z})$ can be modeled as parametric distributions, e.g., $N(\mu_{\mathbf{X}|\mathbf{Z}}(\mathbf{Z}), \Sigma_{\mathbf{X}|\mathbf{Z}}(\mathbf{Z}))$.

An example neural network architecture for deep generative model from Radford (2015).

References I

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- Hyvärinen, A. and Oja, E. (2000). Independent component analysis: algorithms and applications. *Neural networks*, 13(4-5):411–430.
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References II

Szeto, R., Sun, X., Lu, K., and Corso, J. J. (2019). A temporally-aware interpolation network for video frame inpainting. *IEEE transactions on pattern analysis and machine intelligence*, 42(5):1053–1068.